

Question 1**Marks**

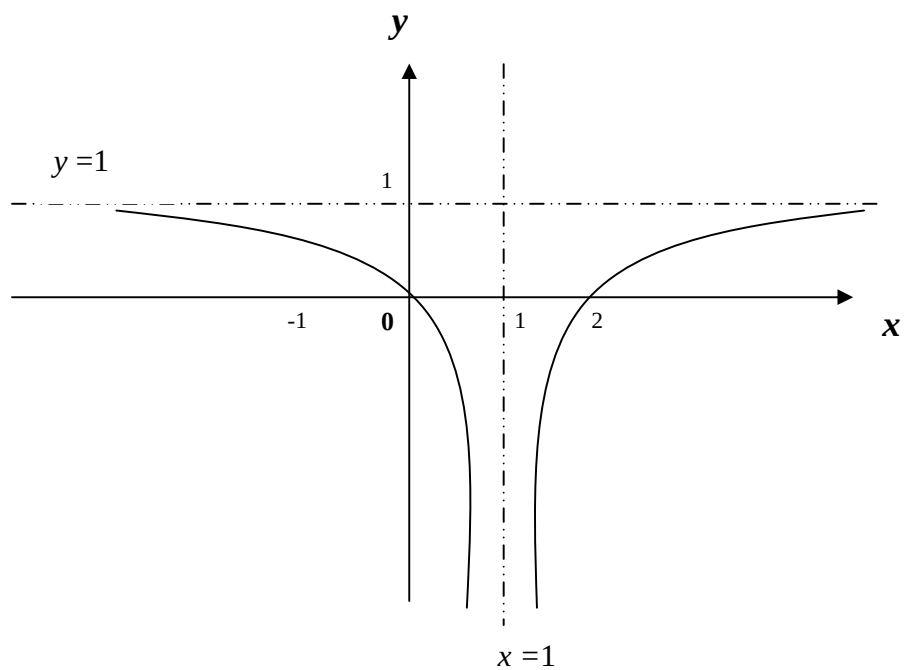
- (a) i) Find $\int \sqrt{e^x} dx$. **1**
- ii) Find $\int \frac{\sqrt{x}}{x-1} dx$. **4**
- iii) Use the substitution $t = \tan \frac{x}{2}$, evaluate $\int_0^{\frac{2\pi}{3}} \frac{1}{5+4\cos x} dx$. **4**
- (b) Consider the hyperbola $\frac{x^2}{4} - \frac{y^2}{3} = 1$.
The point $P(4, 3)$ lies on the hyperbola. The normal at P to the hyperbola meets the x -axis at N . The vertical line through P meets the asymptote in the first quadrant at L .
- i) Sketch the hyperbola, clearly showing all the above information. **2**
- ii) Show that the equation of the normal to the hyperbola at P is $x + y = 7$. **2**
- iii) Show that LN is perpendicular to OL . **2**

Question 2 (Start a new Page)

- (a) Consider the ellipse $\frac{x^2}{16} + \frac{y^2}{12} = 1$.
- i) Sketch the ellipse, clearly showing the directrices and foci. **3**
- ii) The tangent and normal to the ellipse at $P(2, 3)$ cut the y -axis at A and B respectively.
Given that the equation of the tangent to the ellipse at P is $x + 2y = 8$.
Find the coordinates of A and B . **2**
- iii) Show that AB subtends a right angle at the focus (S) of the ellipse. **2**
- iv) Give the reason why the points A, P, S and B are concyclic. **1**
- (b) The graph on the next page shows a function which has x -intercepts at $x = 0$ and $x = 2$. There is a vertical asymptote at $x = 1$ and a horizontal asymptote of $y = 1$.
- Without using calculus, sketch the following graphs on the ANSWER sheet provided at the end of this question paper, clearly showing any asymptotes and intercepts.

Question 2(b) cont'd**Marks**

- | | | |
|------|----------------------|----------|
| i) | $y = f(x) $ | 1 |
| ii) | $y = f(x-1)$ | 1 |
| iii) | $y^2 = f(x)$ | 2 |
| iv) | $y = \tan^{-1} f(x)$ | 3 |

**Question 3 (Start a new Page)**

- | | | |
|--------|--|----------|
| (a) i) | Show that $\sqrt{\frac{8-x}{x}} = \frac{4-x}{\sqrt{8x-x^2}} + \frac{4}{\sqrt{8x-x^2}}$. | 1 |
| ii) | Hence evaluate $\int_0^2 \sqrt{\frac{8-x}{x}} dx$. | 5 |

Question 3 cont'd**Marks**

- (b) $P(a \cos \theta, b \sin \theta)$ is a point in the first quadrant on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and $Q(a \sec \theta, b \tan \theta)$ is a point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, where $a > b > 0$.
- i) Sketch the ellipse, the hyperbola and their common auxiliary circles $x^2 + y^2 = a^2$ and $x^2 + y^2 = b^2$ on the same diagram, showing the angle θ and the related points P and Q . Show clearly how the positions of P and Q are determined by the value of θ , where $0 < \theta < \frac{\pi}{2}$. **3**
- ii) Prove that the tangent to the ellipse at P is $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$. **2**
- iii) The tangent to the hyperbola at Q and the tangent to the ellipse at P meet at T . **4**
 Given that the equation of the tangent to the hyperbola at Q is $\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$.
 Find the coordinates of T .

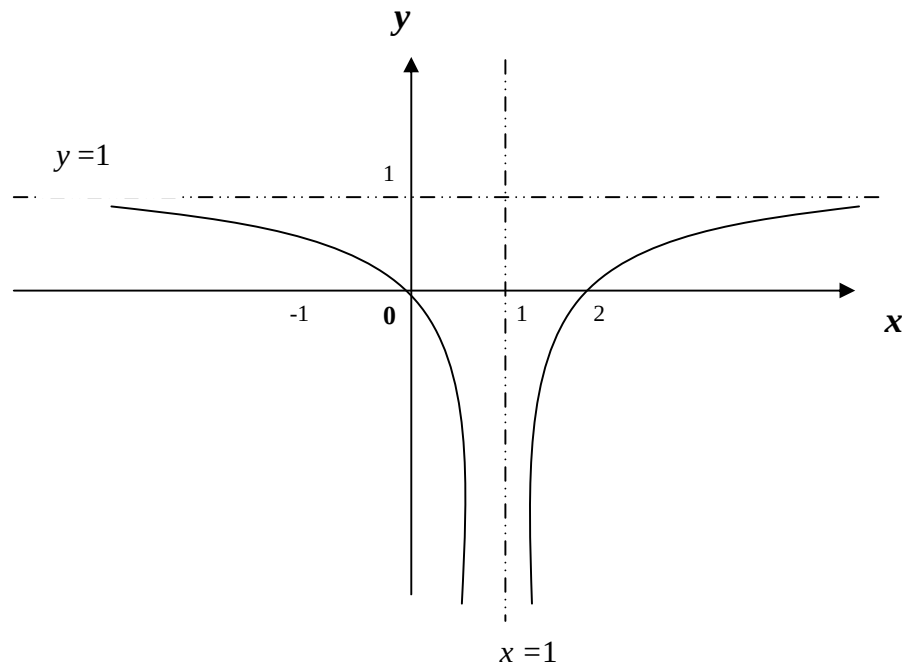
Question 4 (Start a new Page)

- (a) i) Given $f(x) = f(a - x)$ and using the substitution $u = a - x$, prove that $\int_0^a x f(x) dx = \frac{a}{2} \int_0^a f(x) dx$. **3**
- ii) Given that $f(x) = \frac{\sin x}{1 + \cos^2 x}$, prove that $f(x) = f(\pi - x)$. **1**
- iii) Hence evaluate $\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$. **3**
- (b) Let $I_n = \int_0^1 x(1 - x^5)^n dx$, where $n \geq 0$ is an integer.
- i) Show that $I_n = \frac{5n}{5n + 2} I_{n-1}$ for $n \geq 1$. **4**
- ii) Show that $I_n = \frac{5^n n!}{2 \times 7 \times 12 \times \cdots \times (5n + 2)}$ for $n \geq 1$. **3**
- iii) Hence evaluate I_4 . **1**

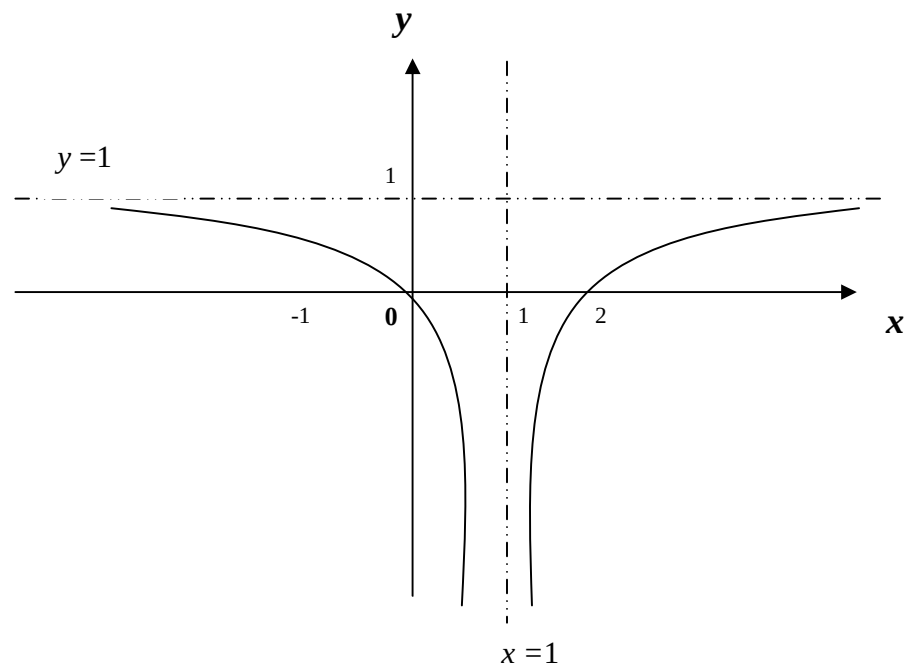
END

Answer sheet for Q2(b)**Student No:** _____

- (i) Sketch $y = |f(x)|$



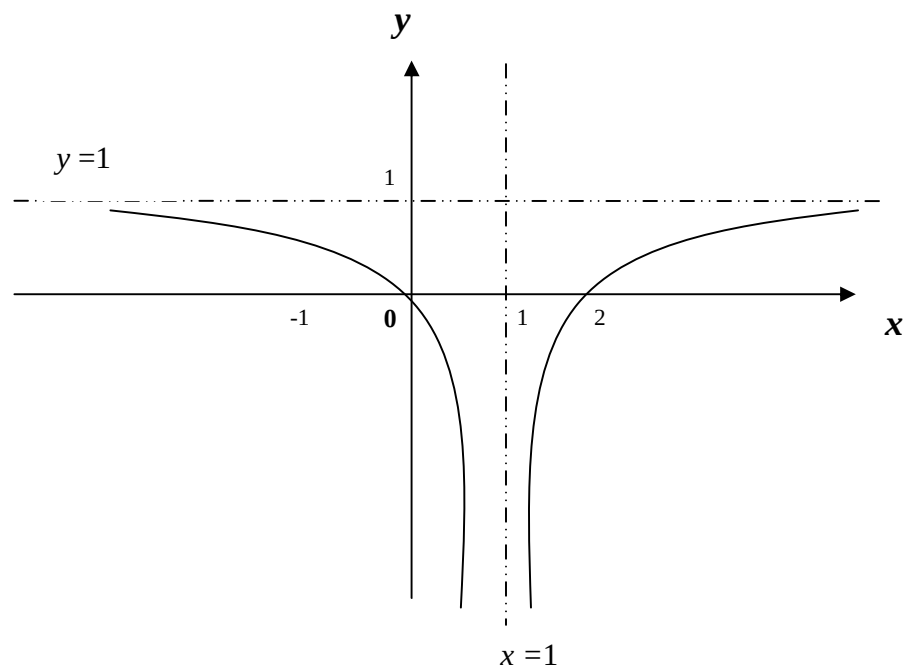
- (ii) Sketch $y = f(x - 1)$

**Turn over for parts(iii) and (iv)**

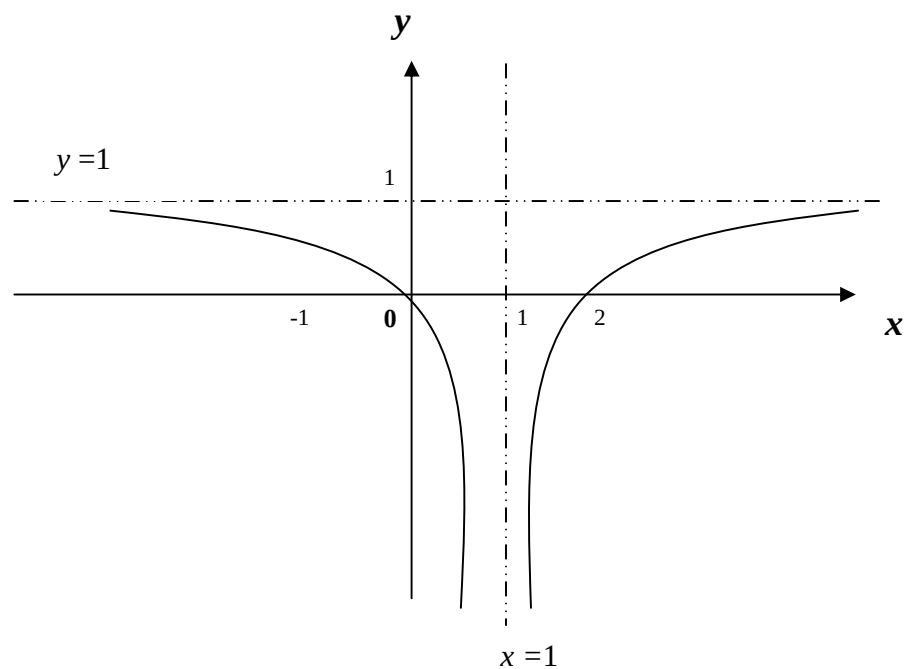
Answer sheet for Q2(b)

Student No. _____

(iii) Sketch $y^2 = f(x)$



(iv) Sketch $y = \tan^{-1} f(x)$



Now attach this sheet with your answer booklet for Question 2.

$$| a.i) \int e^{\frac{x}{2}} dx = 2e^{\frac{x}{2}} + c \quad \#$$

$$ii) 2 \int \frac{u^2 du}{u^2 - 1} \quad \begin{array}{l} \sqrt{x} = u \\ x = u^2 \\ dx = 2u du \end{array}$$

$$= 2 \int \frac{u^2 - 1 + 1 du}{(u^2 - 1)}$$

$$= 2 \int du + 2 \int \frac{du}{u^2 - 1}$$

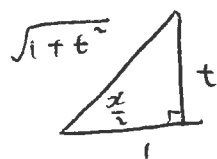
$$= 2u + \int \frac{du}{u-1} - \int \frac{du}{u+1}$$

$$= 2u + \ln\left(\frac{u-1}{u+1}\right) + c$$

$$= 2\sqrt{x} + \ln\left(\frac{\sqrt{x}-1}{\sqrt{x}+1}\right) + c \quad \#$$

$$iii) \int_0^{\frac{2\pi}{3}} \frac{dx}{5+4\cos x} \quad t = \tan \frac{x}{2}$$

$$dx = \frac{2}{1+t^2}$$



$$= \int_0^{\sqrt{3}} \frac{\frac{2}{1+t^2} dt}{5 + 4 \frac{(1-t^2)}{1+t^2}}$$

$$= \int_0^{\sqrt{3}} \frac{2 dt}{5 + 5t^2 + 4 - 4t^2}$$

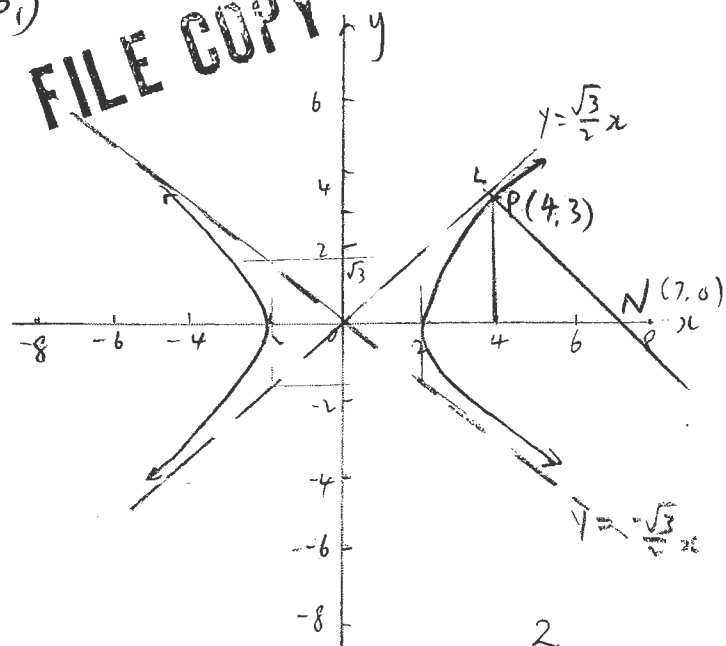
$$= \int_0^{\sqrt{3}} \frac{2 dt}{3 + t^2}$$

$$= \frac{2}{3} \left[\tan^{-1} \frac{t}{\sqrt{3}} \right]_0^{\sqrt{3}}$$

$$= \frac{2}{3} \tan^{-1} \frac{1}{\sqrt{3}} = \frac{2}{3} \times \frac{\pi}{6} = \left(\frac{\pi}{9} \right) \quad |$$

b.i)

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$$ii) \frac{x^2}{4} - \frac{y^2}{3} = 1$$

$$\frac{x}{4} - \frac{yy'}{3} = 0$$

$$\frac{x}{4} = \frac{yy'}{3}$$

$$y' = \frac{x}{4} \cdot \frac{3}{y} \Big|_{(4,3)} = 1$$

$$m_{\perp} = -1$$

$$\text{Eq of Normal at P: } y - 3 = -(x - 4)$$

$$x + y = 7 \quad \#$$

$$iii) \text{ To find L: Eq of asympt. } y = \frac{\sqrt{3}}{2}x$$

$$x = 4, \quad y = \frac{\sqrt{3}}{2} \cdot 4 = 2\sqrt{3}$$

$$\therefore L = (4, 2\sqrt{3})$$

$$\text{To find N: } y = 0, \quad x = 7 \quad N = (7, 0)$$

$$m(LN) = \frac{2\sqrt{3}}{-3} = -\frac{2}{\sqrt{3}}, \quad m(OL) = \frac{2\sqrt{3}}{4} = \frac{\sqrt{3}}{2}$$

$$\therefore m(LN) \times m(OL) = -1$$

$$LN \perp OL \quad \#$$

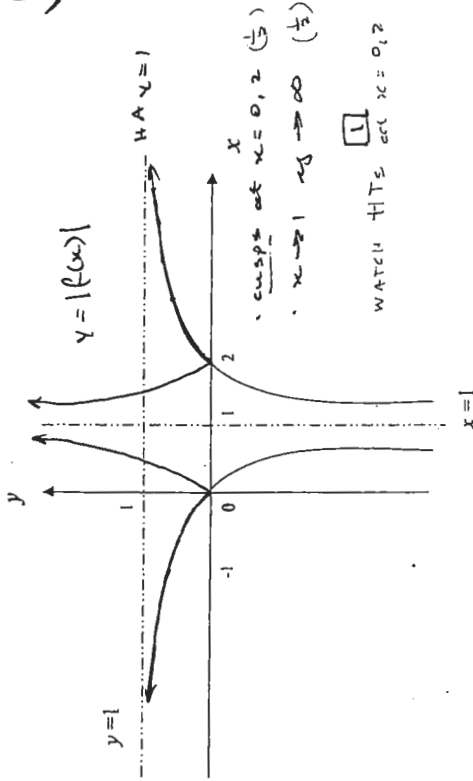
Y12 ASSESSMENT TASK 2, TERM 1 2010
MATH. EXT 2.

MATHEMATICS Extension 2: Question....2

MATHEMATICS Extension 2: Question.....2		
Suggested Solutions	Marks	Marker's Comments
Q2(a)(i) $\frac{x^2}{16} + \frac{y^2}{12} = 1$ (i) $a^2 = 16 \therefore a = 4$ $\therefore e^2 = 1 - \frac{b^2}{a^2} = 1 - \frac{12}{16} = \frac{1}{4}$ $b^2 = 12$ $b = 2\sqrt{3}$ $\therefore e = \frac{1}{2}$ ($0 < e < 1$) $\therefore$ focii $(\pm ae, 0) = (\pm 2, 0)$ Directrices $x = \pm a/e = \pm 4 \div \frac{1}{2} = \pm 8$ NTS [3]		1 For $e = \frac{1}{2}$ $\frac{1}{2}$ For $S(2, 0)$ $\frac{1}{2}$ For $x = 8$ 1 For Ellipse shape
(ii) $x + 2y = 8$ For A: $x = 0 \therefore y = 4$ $A = (0, 4)$ Gradient of tangent $m_T = -\frac{1}{2}$ " " normal $m_N = +2$ $\therefore$ Eqn. of normal $y - 3 = 2(x - 2)$ $y = 2x - 1$ For B: $x = 0 \therefore y = -1$ $B = (0, -1)$ [2]	$\frac{1}{2}$ $\frac{1}{2}$ 1	$\frac{1}{2}$ For $(0, 4)$ $\frac{1}{2}$ For $m_N = 2$ 1 For $(0, -1)$
(iii) $A(0, -4)$ $S(2, 0)$ $B(0, -1)$ $m_{AS} = \frac{4 - 0}{0 - 2} = -2$ $m_{SB} = \frac{-1 - 0}{0 - 2} = +\frac{1}{2}$ $\therefore m_{AS} \times m_{SB} = -2 \times \frac{1}{2} = -1$ $\therefore AS \perp SB \therefore \angle ASB = 90^\circ$ i.e. AB subtends a right angle at S. [2]		$\frac{1}{2}$ For $m_{AS} = -2$ $\frac{1}{2}$ For $m_{SB} = \frac{1}{2}$ $\frac{1}{2}$ For showing $\frac{1}{2}$ For conclusion
(iv) $\angle APB = 90^\circ$ (Tangent and normal at 90°) ✓ $\angle ASB = 90^\circ$ (shown above) $\therefore$ A, P, S and B are concyclic points as interval AB subtends equal angles at 2 points on the same side of it, then endpoints and the 2 points are concyclic • could reason using semicircles, diameter + reason. [1]		1 For 'correct reason(s)'

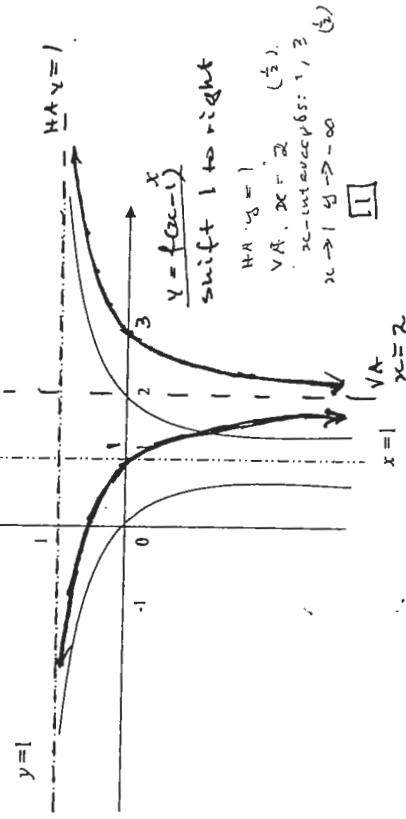
(b) (i)

Sketch $y = |f(x)|$



(ii)

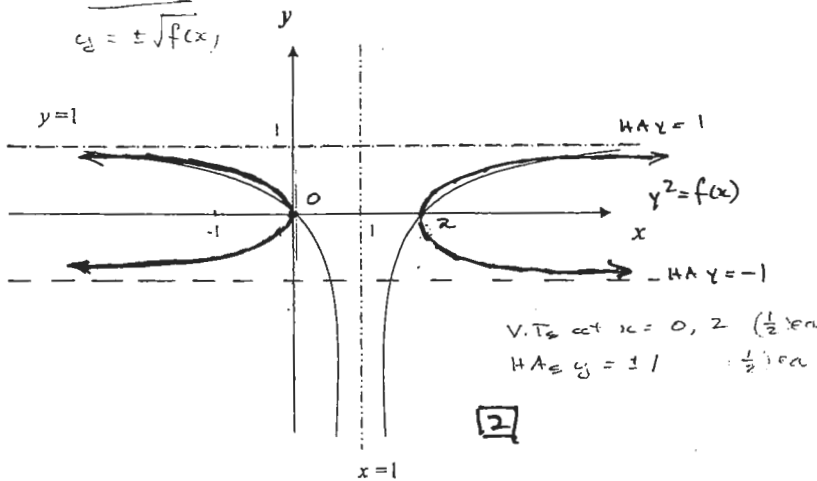
Sketch $y = f(x-1)$



(iii)

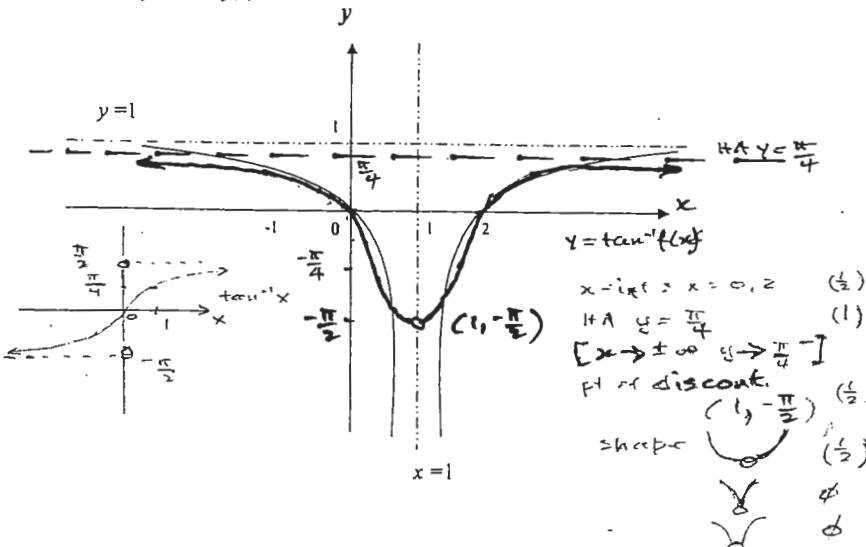
Sketch $y^2 = f(x)$

$$y = \pm \sqrt{f(x)}$$



(iv)

Sketch $y = \tan^{-1} f(x)$



$$3 \text{ a)} \sqrt{\frac{8-x}{x}} = \sqrt{\frac{8-x}{x}} \cdot \sqrt{\frac{8-x}{x}} = \frac{8-x}{\sqrt{(8-x)x}}$$

$$= \frac{4-x}{\sqrt{8x-x^2}} + \frac{4}{\sqrt{8x-x^2}} \quad \#$$

$$\therefore \int_0^2 \sqrt{\frac{8-x}{x}} dx = \int_0^2 \frac{4-x}{\sqrt{8x-x^2}} dx + 4 \int_0^2 \frac{dx}{\sqrt{8x-x^2}}$$

$$\int_0^2 \frac{2(4-x)dx}{\sqrt{8x-x^2}} = \frac{1}{2} \left[\left(\sqrt{8x-x^2} \right) \right]_0^2 = \sqrt{16-4}$$

$$= \sqrt{12} = 2\sqrt{3} \quad \#$$

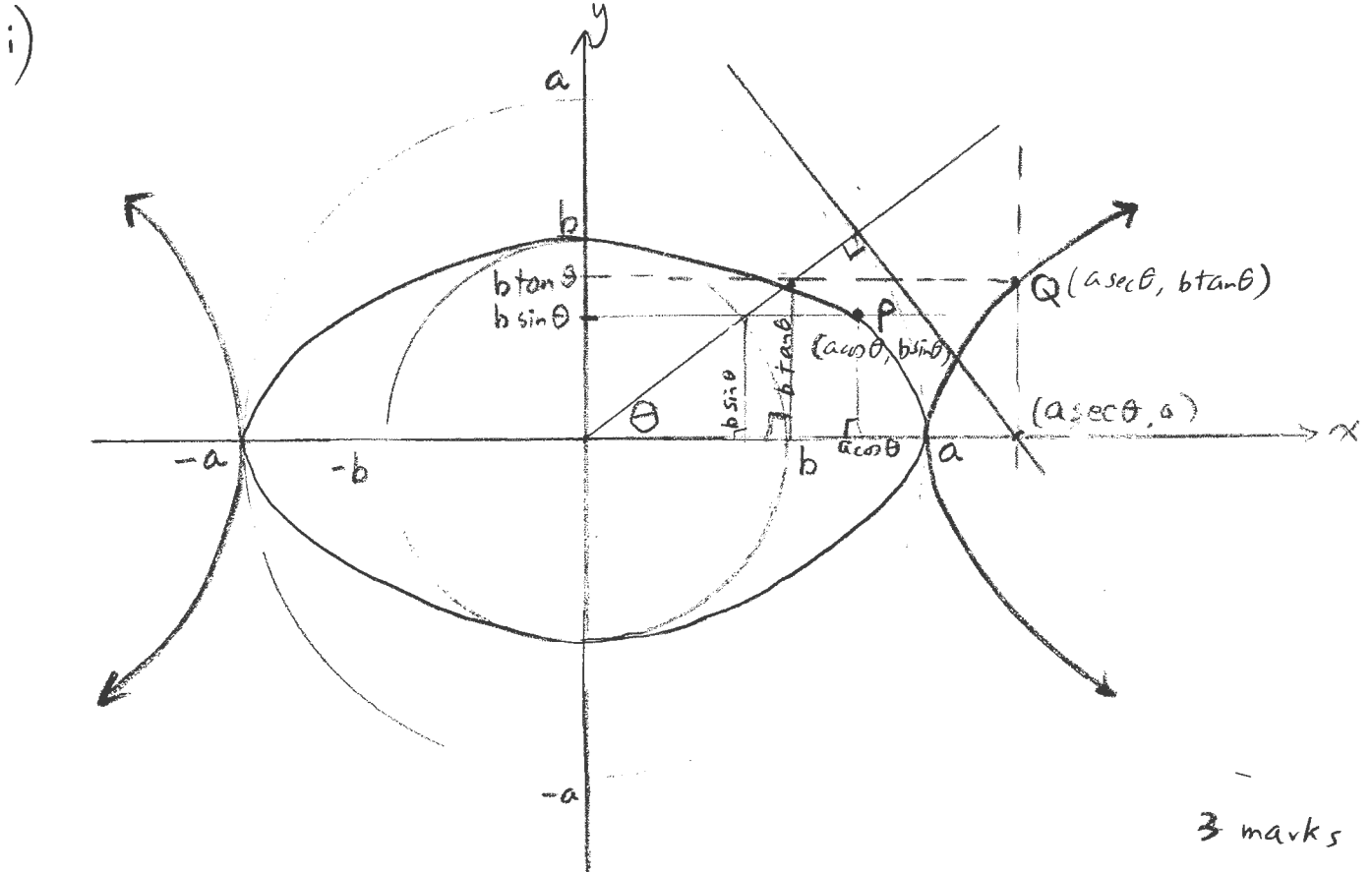
$$4 \int_0^2 \frac{dx}{\sqrt{8x-x^2}} = 4 \int_0^2 \frac{dx}{\sqrt{-(x^2-8x+16-16)}}$$

$$= 4 \int_0^2 \frac{dx}{16-(x-4)^2} = 4 \left[\sin^{-1} \left(\frac{x-4}{4} \right) \right]_0^2$$

$$= 4 \sin^{-1} \left(-\frac{1}{2} \right) - \sin^{-1}(-1)$$

$$= 4 \left(-\frac{\pi}{6} + \frac{\pi}{2} \right) = \frac{4\pi}{3} \quad \#$$

$$\text{Ans } 2\sqrt{3} + \frac{4\pi}{3} \quad \#$$



ii) $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = -\frac{b \cos \theta}{a \sin \theta}$

Eq of tangent at P: $y - b \sin \theta = -\frac{b \cos \theta}{a \sin \theta} (x - a)$
 $x b \cos \theta + y a \sin \theta = ab (\cancel{\cos^2 \theta} + \cancel{\sin^2 \theta})$
 $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$

iii) $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$ ① $x \sec \theta + \frac{y \tan \theta}{b} = \sec \theta$ ③

$\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$ ②

② + ③ $\frac{x}{a}(1 + \cancel{\sec \theta}) = 1 + \cancel{\sec \theta}$ $\therefore x = a$ #

Sub $x = a$ into ① $\sec \theta + \frac{y \sin \theta}{b} = 1$

$\frac{y \sin \theta}{b} = 1 - \cos \theta$

$y = \frac{b}{\sin \theta} (1 - \cos \theta)$ #

MATHEMATICS Extension 2: Question ... 4

Suggested Solutions

Marks

Marker's Comments

$$\begin{aligned}
 \text{a) (i)} \quad f(x) &= f(a-x) \quad u = a-x \\
 \int_0^a x f(x) dx & \quad x = a-u \\
 & \quad x=0 \quad u=a \\
 & \quad x=a \quad u=0 \\
 & \quad dx = -du \\
 &= \int_a^0 (a-u) f(a-u) (-du) \\
 &= \int_0^a (a-u) f(u) du \\
 &= a \int_0^a f(u) du - \int_0^a u f(u) du \\
 & \quad \text{change variable} \\
 \int_0^a x f(x) dx &= a \int_0^a f(x) dx - \int_0^a x f(x) dx \\
 2 \int_0^a x f(x) dx &= a \int_0^a f(x) dx \\
 \int_0^a x f(x) dx &= \frac{a}{2} \int_0^a f(x) dx
 \end{aligned}$$

① correct substitution

① change of variable

split integral
① change to $f(x)$ to give answer

$$\text{(ii)} \quad \frac{\sin(\pi-x)}{1 + [\cos(\pi-x)]^2} = \frac{\sin x}{1 + (-\cos x)^2}$$

$$\begin{aligned}
 \text{because} \quad \sin(\pi-x) &= \sin x \\
 \cos(\pi-x) &= -\cos x
 \end{aligned}$$

$$\therefore f(\pi-x) = \frac{\sin x}{1 + \cos^2 x} = f(x)$$

$$\text{(iii)} \quad \int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx = \frac{\pi}{2} \int_0^\pi \frac{\sin x}{1 + \cos^2 x} dx$$

must show
① show
 $\sin(\pi-x) = \sin x$
 $\cos(\pi-x) = -\cos x$

change
① integral

① integrate

$$\text{as } f(\pi-x) = f(x) \quad = -\frac{\pi}{2} \left[\tan^{-1}(\cos x) \right]_0^\pi$$

$$= -\frac{\pi}{2} \tan^{-1}(\cos \pi - \cos 0)$$

$$= -\frac{\pi}{2} (\tan^{-1}(-1) - \tan^{-1}(1))$$

$$= -\frac{\pi}{2} \left(-\frac{\pi}{4} - \frac{\pi}{4} \right)$$

$$= \frac{\pi^2}{4}$$

① answer

MATHEMATICS Extension 2: Question 4...

Suggested Solutions	Marks	Marker's Comments
(i) $I_n = \int_0^1 x(1-x^5)^n dx$ $= \left[\frac{x^2}{2} (1-x^5) \right]_0^1 - \int_0^1 \frac{x^2}{2} n (1-x^5)^{n-1} (-5x^4) dx$ $= (0-0) + \frac{5n}{2} \int_0^1 x^6 (1-x^5)^{n-1} dx$ $= \frac{5n}{2} \int_0^1 x(x^5 - 1 + 1)(1-x^5)^{n-1} dx$ $= \frac{5n}{2} \int_0^1 x(1-x^5)^{n-1} dx - \frac{5n}{2} \int_0^1 x(1-x^5)^n dx$ $I_n = \frac{5n}{2} I_{n-1} - \frac{5n}{2} I_n$ $I_n \left[1 + \frac{5n}{2} \right] = \frac{5n}{2} I_{n-1}$ $I_n = \left(\frac{5n}{2+5n} \right) I_{n-1}$		① Integrate by parts ① simplify ① separate integrals ① complete solution
(ii) $I_n = \frac{5n}{2+5n} I_{n-1} \quad n \geq 1$ $I_1 = \frac{5}{7} I_0 \quad I_0 = \int_0^1 x(1-x^5)^0 dx$ $I_2 = \frac{10}{12} I_1 = \int_0^1 x dx$ $I_3 = \frac{15}{17} I_2 = \left[\frac{x^2}{2} \right]_0^1 = \frac{1}{2}$ $I_n = \frac{5n}{5n+2} \cdot \frac{5(n-1)}{5(n-1)+2} \cdot \frac{5(n-2)}{5(n-2)+2} \cdots \frac{5 \times 3}{5 \times 3 + 2} \cdot \frac{5 \times 2}{5 \times 2 + 2} \times I_0$ $= \frac{5^n (n)(n-1)(n-2) \cdots 3 \times 2 \times 1}{(5n+2)(5n-3)(5n-8) \cdots 17 \times 12 \times 7}$ $= \frac{5^n n!}{2 \times 7 \times 12 \times \cdots (5n+2)}$ $I_4 = \frac{5^4 \times 4!}{2 \times 7 \times 12 \times 17 \times 22} = \frac{15000}{62832}$ $= \frac{625}{2618}$		② must show series at both ends including I_0 ① calculate I_0 ③ ② must show 5 outside each term to make 5^n and $n(n-1)(n-2) \cdots 3 \times 2 \times 1$ ① correct answer only.